BATCH LEARNING

- The conventional machine learning is rooted in the *statistical learning theory* and is sometimes referred to as the *batch learning scenario*:
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Advanced Machine Learning
Introduction to Online Learning

Learning goals

- Understand the difference between batch and online learning
- Know the basic and the extended learning protocol in online learning
- Know how performance is measured in online learning

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ONLINE LEARNING

- However, many real-world problems are *dynamic* with the following aspects:
  - *Sequential order* — data is generated only bit by bit;
  - *On-the-fly decisions* — decisions or predictions have to be made during the data generating process;
  - *Unforeseeable consequences* — decisions can have a drastic influence on the data generating process;
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- Characteristically: In the online learning scenario the training phase and the testing phase are **interleaved**.

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⇒ The learner and the environment are alternately performing their actions.

THE BASIC ONLINE LEARNING PROTOCOL

Formally, an online learning problem consists of:

- a learner (forecaster, agent resp. decision maker), an environment (user resp. adversary, system resp. nature),
- time steps $1, 2, \ldots, T$ (may be infinite),
- available actions $\mathcal{A}$ for the learner (may be infinite),
- environmental data space $\mathcal{Z}$,
- a loss function $L : \mathcal{A} \times \mathcal{Z} \to \mathbb{R}$.

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Typically $\mathcal{A} = \mathcal{Z} = \mathcal{Y}$, so that

- the learner's chosen action $a_t = \hat{y}_t$ corresponds to a prediction,
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THE EXTENDED ONLINE LEARNING PROTOCOL

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- Typically $\mathcal{A} = \mathcal{Y}$ and $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, so that
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DATA GENERATION IN ONLINE LEARNING PROTOCOL

- In many applications, the environmental data consists of two parts $z_t = (z_t^{(1)}, z_t^{(2)})$, where the first part of the data, $z_t^{(1)}$, is revealed to the learner **before** the action is made. After the learner carries out its action, the remaining part of the environmental data is revealed, that is, $z_t^{(2)}$.
- In particular, the online learning problem is now as follows in each time step:
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- In particular, the environmental data are not necessarily generated by a probability distribution!
- This also covers the area of *adversarial learning*: the data can even be generated by an adversary trying to fool the learner.
- However, the online learning setting can of course also be considered in a statistical setting.

ONLINE LEARNING: REQUIREMENTS

- The dynamical aspects have to be incorporated for the design of efficient learning algorithms.
- The online learner has to cope with the sequential availability of the data and to cope with time as well as computational constraints.
- Roughly speaking, one seeks to construct an online learning algorithm which is adaptive to the environment and allows incremental as well as preferably cheap updates over time.

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- Although consideration of time and memory constraints is important for practical purposes, we will only implicitly consider these constraints in this lecture.
- We will mainly focus our theoretical analysis on the performance of the learner in terms of its (cumulative) loss, which, however, will usually ignore computational aspects of the learner.

MEASURE OF QUALITY IN ONLINE LEARNING

- In order to measure the quality of an online learner one can compute the difference between the cumulative loss of the learner and the cumulative loss by taking some competing action $a \in \mathcal{A}$:

$$R_T(a) = \sum_{t=1}^{T} L(a_t, z_t) - \sum_{t=1}^{T} L(a, z_t).$$

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- Formally, the following should hold

$$R_T = o(T).$$

Interpretation: The average regret per time step (or per example) goes to zero:

$$\frac{1}{T}\Big(\sum_{t=1}^{T} L(a_t, z_t) - \inf_{a \in \mathcal{A}} \sum_{t=1}^{T} L(a, z_t)\Big) = \frac{R_T}{T} = o(1).$$

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- One might ask why one compares only with a fixed best action in hindsight, say a^*, instead of a sequence of actions $a_1^*, a_2^*, \ldots, a_T^*$?
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